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Perspective

Perspective: An International Fludarabine Shortage: Supply Chain Issues Impacting Transplantation and Immune Effector Cell Therapy Delivery

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Host immune depletion has been recognized as a necessary step for successful adoptive immune cell transfer in both the autologous and allogeneic settings. The chemotherapy agent fludarabine as an immune suppressive agent has a central role in multiple conditioning regimens for both transplantation and immune effector cell therapies. With the recent and sudden recognition of an imminent worldwide fludarabine shortage, novel approaches to overcome supply chain disruption are needed, including exploration of alternative therapies. The fludarabine shortage has highlighted the need to prioritize the development of institutional algorithms for maintaining ongoing clinical trials and standard of care procedures in the setting of critical drug shortages.

Host immune depletion has been recognized as a necessary step for successful adoptive immune cell transfer in both the autologous and allogeneic settings. As such, we are notifying our community that we face an imminent worldwide fludarabine shortage and herein discuss alternative therapies and recommend prioritizing the development of institutional algorithms for fludarabine use in ongoing clinical trials and standard of care procedures.

Successful allogeneic hematopoietic cell transplantation (HCT) depends on host immune depletion to eliminate the risk of allograft rejection, highlighted by early experience in aplastic anemia, with heavily transfused patients found to have higher rates of graft rejection [1]. Ex vivo T cell-depleted haploidentical allogeneic transplantation with suboptimal host immunosuppression often fails owing to graft rejection [2]. Nonmyeloablative allogeneic transplantation with minimal intensive regimens has been augmented by deeper host depletion; specifically, the addition of fludarabine to 2 Gy total body irradiation (TBI) decreased host allograft rejection from 17% to 3%, particularly in the unrelated population [3].

Host lymphodepletion for autologous adoptive immunotherapy therapy has long been recognized as one of the key central requirements for successful intervention, dating to the original murine model observations demonstrating that cyclophosphamide-facilitated adoptive immunotherapy of a target malignancy depends on the elimination of tumor-induced suppressor cells [4]. Work by Rosenberg and others confirmed that adoptive immunotherapy of cancer with tumor-infiltrating lymphocytes (TILs) was only possible with deep immune suppression in which dosage-escalated cyclophosphamide (60 mg/kg for 2 days) in conjunction with fludarabine was administered [5–7]. Further exploration of optimizing adoptive immunotherapy revealed that successful preinfusion lymphoablation was achieved by removing consumptive cytokine sinks, eradicating inhibitory effects of T regulatory cells and enhancing activation of antigen-presenting cells, achieved by inducing tumor apoptosis/necrosis, resulting in the uptake and presentation of tumor antigens and activation of antigen-presenting cells, enhancing stimulation of adoptively transferred T cells [8].

Chimeric antigen receptor T cell (CAR-T) therapy has rapidly become a widely used standard of care treatment option for patients with advanced lymphoid malignancies, with anticipated expansion of these therapies into many other liquid and solid cancers [9]. The critical need for lymphodepleting chemotherapy (LDC) for CAR-T success is well recognized. Early studies (initiated in 2013) of pediatric acute

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lymphoblastic leukemia demonstrated that an increased cyclophosphamide dose (3 g/M^2 versus $\leq$1.5 g/M^2) was associated with improved responses and better CAR T cell expansion; interestingly, only 6 of the 25 subjects received additional fludarabine [10]. Subsequently, Turtle et al. [11] clearly demonstrated that the addition of fludarabine to cyclophosphamide was associated with greater cellular expansion and persistence of the CAR T cells. The same group using a fludarabine-based regimen further confirmed the benefit of cyclophosphamide dose escalation, where they felt that the benefit of the high-intensity LDC was to generate a more favorable cytokine profile promoting better CAR T cell expansion, contributing to improved progression-free survival, and not just a product of an antimalignancy benefit [12]. Multiple other agents have been assessed for their ability to be effective LDC agents, including TBI, etoposide, carboplatin/gemcitabine, pentostatin, and bendamustine [13]. The greatest experience outside of fludarabine and cyclophosphamide combination has been with bendamustine [13,14]. A recent comparative study of bendamustine and fludarabine/cyclophosphamide LDC suggested equivalence, but this has been relatively limited to CAR-T therapies constructed with the 4-1BB T cell activation domain [14]. However, it is important to recognize that among the >5000 patients now reported in the Chemotherapy Immunotherapy Data Resource of the Center for International Blood and Marrow Transplant Research, mostly recipients of commercial CAR-T products, the vast majority received a combination of fludarabine- and cyclophosphamide-based LDC following manufacturer's guidelines (Table 1), with occasional minor variations dependent on the product used [15,16]. Finally, confirming the need for further exploration of LDC, recent studies suggest that suboptimal fludarabine exposure, defined as a concentration-time curve, reveal that an area under the curve value of <13.8 to 14 mg $\times$ h/L was associated with higher relapse rates compared with more optimal exposure [17,18].

Thus, combinations of cyclophosphamide and fludarabine remain the backbone of LDC, with recognition that future approvals will dramatically increase these demands. Multiple myeloma (MM) is the second most frequent hematologic malignancy, and 2 CAR-T products are currently approved for patients with relapsed, refractory MM [19,20]. Recognizing that the median survival with this disease is nearly 10 years, it is likely inevitable that most of the thousands of individuals with MM will become candidates for this treatment during their lifetime. Similarly, and importantly, the results of the recent ZUMA-7 [21] and TRANSFORM [1,22] clinical trials for diffuse large B cell lymphoma (DLBCL) have now made CAR-T a standard of care option for early relapsing DLBCL within 1 year of initial therapy, which is estimated to increase the number of eligible candidates for CAR-T significantly [23].

It is in this background that our community has been faced with the relative sudden notification that there is a worldwide fludarabine shortage [24]. Fludarabine has been identified in the drug shortage trackers since December 2021. There was transient stabilization in March 2022, but again, the supply from Hope Sellers has been depleted, with no estimated time of resolution. The manufacturers have not provided a reason for the shortage, and only limited products are available, with one vendor imposing an approximate 1000% cost increase for vial acquisition. Considerations for dealing with shortages at a local level already have been implemented at some institutions (aside from attempting to obtain supply from neighboring, and often-competing, hospital systems), including (1) pharmacy dose banding and rounding down to save vials, even if a >5% reduction was required; (2) administering all dosing of fludarabine based not on actual body weight but on adjusted body weight; and (3) switching the billing of fludarabine from single-dose vials to billing by dose delivery. This latter approach will allow split vial delivery between patients and asking clinical teams to schedule fludarabine requiring

Table 1
FDA-Approved Commercial CAR-T Cell Products

Product	Disease	Recommended LDC (FDA Label)	Administration of LDC and CAR-T Therapy
Axicabtagene ciloleucil	Adult large cell lymphoma, second line, third line	CTX 500 mg/m^2 Flu 30 mg/m^2	Days -5, -4, and -3
Axicabtagene ciloleucil	Adult follicular lymphoma	CTX 500 mg/m^2 Flu 30 mg/m^2	Days -5, -4, and -3
Tisagenlecleucel	Pediatric/young adult ALL	CTX 500 mg/m^2 $\times$ 2 days (start with dose 1 of Flu) Flu 30 mg/m^2 $\times$ 4 days	LDC between days -14 and -2
Tisagenlecleucel	Adult large cell lymphoma, third line	CTX 250 mg/m^2 $\times$ 3 days Flu 25 mg/m^2 $\times$ 3 days Alternate LDC: bendamustine 90 mg/m^2 $\times$ 2 days	Administer LDC between days -11 and -2
Tisagenlecleucel	Adult follicular lymphoma	CTX 250 mg/m^2 $\times$ 3 days Flu 25 mg/m^2 $\times$ 3 days Alternate LDC: bendamustine 90 mg/m^2 $\times$ 2 days	Administer LDC between days -6 and -2
Lisocabtagene maraleucel	Adult large cell lymphoma, second line, third line	CTX 300 mg/m2 $\times$ 3 days Flu 30 mg/m2 $\times$ 3 days	Administer LDC between days -7 and -2
Brexucabtagene autoleucel	Mantle cell lymphoma	CTX 500 mg/m^2 $\times$ 1 day Flu 30 mg/m^2	Days -5, -4, and -3
Brexucabtagene autoleucel	Adult ALL	CTX 900 mg/m^2 Flu 25 mg/m^2 $\times$ 3 days	Flu on days -4,-3, and -2; CTX on day -2
Idecabtagene vicleucel	MM, beyond fourth line	CTX 300 mg/m^2 $\times$ 3 days Flu 30 mg/m^2 $\times$ 3 days	Days -4, -3, and -2
Ciltacabtagene autoleucel	MM, beyond fourth line	CTX 300 mg/m^2 $\times$ 3 days Flu 30 mg/m^2 $\times$ 3 days	Administer LDC between days -6 and -2

CTX indicates cyclophosphamide; Flu, fludarabine.

patients on the same days so that drug wastage can be limited. Finally, there is evidence to support the storage of reconstituted fludarabine at 2 °C to 8 °C, with stability identified by HPLC at 14 to 21 days, suggesting that reconstituted product may be able to be saved short term for patient use [25,26].

From a clinical perspective and the transplantation and cell therapy arena, fludarabine has become an essential agent for conditioning. The clinical consequence of removing or replacing fludarabine from lymphodepletion prior to CAR-T cell therapy is unknown but is potentially life-altering for a treatment given with curative intent. The even greater need for and usage of fludarabine remains in allogeneic transplantation, where conventional and reduced-intensity myeloablative regimens using fludarabine are some of the most widely used conditioning regimens in the United States (fludarabine/melphalan; busulfan2/fludarabine; busulfan4/fludarabine), as are many haploidentical mismatched conditioning strategies. As stated above, fludarabine and cyclophosphamide have been the central regimens for CAR-T therapy, both commercial and investigational. In the investigational arena, the use of fludarabine as the base of LDC extends to other cell populations, including natural killer cell and TIL cell products.

In the event that the fludarabine shortage is persistent, recognizing that 3 of the 5 manufacturers still cannot estimate a release date [24], it becomes necessary for centers to establish algorithms for management now. Substitution of such agents as bendamustine and cladribine can be considered, particularly for application to 4-1BB (CD137) activating domain CAR-T products. In our recent comparison of bendamustine with cyclophosphamide and fludarabine, clinical outcomes appeared similar, but preliminary examination of the depth of lymphodepletion suggest that bendamustine did not reach the same level of depletion as the 2-drug regimen [14]. Regarding cladribine, this agent has similar pharmacologic properties as fludarabine and is now accepted as an effective immune suppressive agent and has been approved by the Food and Drug Administration (FDA) for the treatment of multiple sclerosis. However, little data on usage are currently available, and cladribine is associated with myelotoxicity and might not be the best agent for CAR-T therapy, with up to 30% of patients reported as having late cytopenias in a reduced-intensity randomized trial comparing fludarabine and cladribine that was halted owing to a recognized potential for early graft failure in the cladribine arm [27].

Finally, another acceptable solution could be the substitution of clofarabine for fludarabine. This agent, which is highly effective in lymphoid and myeloid malignancies, is in greater supply and for many situations, particularly in the HCT setting, could be an acceptable interchangeable agent, while recognizing that it may carry an increased risk of infection [28–31]. However, also in HCT, where reimbursement is often under a case rate for private payers or under a diagnosis-related group for Medicare reimbursement, one must be aware that the cost of clofarabine is approximately 10-fold higher than that of fludarabine.

Thus, for allogeneic transplantation, substitution of busulfan and cyclophosphamide for busulfan/fludarabine-based myeloablative regimens and clofarabine/melphalan for fludarabine/melphalan could be pursued. However, for immune effector cell therapy, the limited available data preclude strong recommendations for substitutes. In clinical trials, where the variable assessed is efficacy of a product, altering the LDC regimen could create a significant confounder and threaten data validity. It is our opinion that to maintain clinical trial standards and data validity, our immune effector community must remain committed to strict adherence to the protocol-defined lymphodepletion in ongoing clinical trials, unless the FDA allows for substitutions to corporate sponsors. Real-world alterations of fludarabine-based LDC can be performed with the commitment that in the future, these subjects treated with modified LDC will be examined retrospectively by the Chemotherapy Immunotherapy Data Resource with the goal of

Table 2
OHSU Institutional Algorithm for Fludarabine Management in Setting of Critical Drug Shortage

1. Adjust dose delivery to avoid usage waste (vial sparing, dose banding, adjusted body weight dosing)
2. Investigational study subjects
a. If the study provides fludarabine: proceed per study protocol with sponsor supply
b. If no standard of care option for CAR-T application (eg, advanced CLL): verify fludarabine supply and proceed with study participation
c. If study allows alternative regimens: proceed with recommended alternative
d. For enrolled patients: determine with study sponsor if rounding down is acceptable for vial sparing
e. If none of the above is available/feasible: STOP enrollment of new patients until fludarabine shortage resolved
3. Standard of care (SOC) alternative regimen recommendations
a. Allogeneic HCT regimen alternatives (currently prioritize allogeneic HCT over CAR-T and leukemia indications, based on patient volumes)
i. Substitute BuFlu or FluMel with →
1. If eligible for MAC
a. BuCy or Clo/Bu
b. CyTBI or busulfan/etoposide (consider for ALL)
2. If eligible for RIC
a. Clo/Mel
b. Bu/Cy with reduced-dose busulfan (12 mg/kg)
3. Delay transplantation (eg, MDS stable on HMAs, AML in CR with maintenance options, multiple URD options)
ii. Haplo regimens (Flu/Cy/TBI + post-transplantation Cy; FluTBI + PTCy)
1. BuCy (use IBW dosing only)
2. Consider a Clo-based regimen
3. Delay haploidentical transplantation
b. CAR-T lymphodepletion conditioning
i. Tisagenlecleucel
1. Bendamustine → adult preferred regimen
2. Cytarabine + etoposide → alternate pediatric ALL regimen (ELIANA trial)
ii. All other SOC CAR-T cell products
1. For ALL and MM: prefer Flu/Cy if available; otherwise bendamustine
2. For all other indications (MCL, FL, and DLBCL): bendamustine the preferred regimen until fludarabine shortage considered resolved
c. Leukemia salvage regimens
i. FLAG or FLAG-IDA salvage: switch entirely to CLAG/CLAG-M/GCLAM (cladribine ± mitoxantrone) → preferred alternative regimens
ii. Alternate salvage: HMA/venetoclax; HMA/gemtuzumab

CLL indicates chronic lymphoblastic leukemia; Bu, busulfan; Mel, melphalan; MAC, myeloablative conditioning; Cy, cyclophosphamide; RIC, reduced-intensity conditioning; Clo, clofarabine; MDS, myelodysplastic syndrome; HMA, hypomethylating agent; AML, acute myelogenous leukemia; CR, complete response; URD, unrelated donor; PTCy, post-transplantation cyclophosphamide; IBW, ideal body weight; ALL, acute lymphoblastic leukemia; MCL, mantle cell lymphoma; FL, follicular lymphoma; FLAG, fludarabine/cytarabine; IDA, idarubicin identifying whether any changes in efficacy or toxicity were encountered.

It is critical that this issue be shared with the membership of ASTCT and other societies focused on the expansion of cellular therapies. One only need review the recent rapid expansion of CAR-T therapy, which has grown from fewer than 200 procedures per year in 2016 to >1700 per year in 2020 (for which data are still incomplete), and consider that since that time, multiple new approved indications, including mantle cell lymphoma, MM, follicular lymphoma, and first relapse DLBCL have occurred, to recognize that we have a mandate as a scientific community to rapidly study alternate LDC approaches given the ongoing supply chain and drug shortages that are now no longer a rare event. Additionally, as a society, we need to make drug shortages, specifically of fludarabine at this time, as one of our highest priorities to address and bring our influence to bear on both industry and governmental agencies to facilitate rapid resolution of the shortage. Although prior shortages have had acceptable alternatives (eg, substitution of normal saline with alternative fluids, shortage of i.v. dexamethasone), the current problem is unique in that we do not have an adequately studied alternative regimen to replace our standard of care fludarabine/cyclophosphamide regimen for CAR-T therapy.

We implore the ASTCT to endorse and treatment centers to implement fludarabine rationing plans that will maximize patient benefit and prioritize clinical trial integrity while advocating with suppliers and regulatory agencies to avoid future drug shortages. Finally, it will be appropriate to establish strict algorithms for managing the limited current supplies to guide an institutional approach to patient care that is transparent and reproducible. We share one such algorithm with the reader, currently in place at Oregon Health Science University and under constant (if not weekly) revision, recognizing that these solutions are not easy and are best made locally and influenced by unique institutional needs (Table 2).

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